

Annotation Transfer Tool

<https://github.com/aiyshwary/Annotation-Transfer-Tool>

Python

Computer Vision

AWS S3

CVAT

PyTorch

OVERVIEW

A smart auto-annotation pipeline that leverages deep image similarity to transfer labels from previously human-annotated images to new unannotated ones, significantly reducing manual labeling effort in computer vision workflows.

IMPACT

Reduced manual annotation effort by automatically transferring labels from existing human-labeled images to new images using deep feature similarity and a modular 8-step pipeline integrated with CVAT and AWS S3.

KEY DETAILS

- Downloads labeled images from AWS S3 and extracts annotations from Datamano JSON format.

- Generates image crops, normalizes labels, and creates masks for each labeled region.

- Uses DeepImageSearch (GPU-accelerated) to identify the most visually similar labeled images for each new image.

- Transfers and corrects annotations from matched images, then exports back to CVAT-compatible JSON.

- Includes normalized and denormalized visualization scripts and an interactive Jupyter demo notebook.